

Repeaterless Low-Swing On-Chip Interconnects

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under the guidance of
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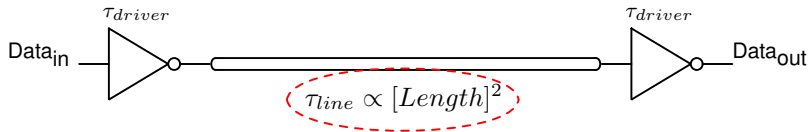


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Introduction

- With technology scaling transistor performance has been improving.
- Interconnect performance has however not been scaling at the same rate. This has led to interconnect performance limiting the performance of digital systems.

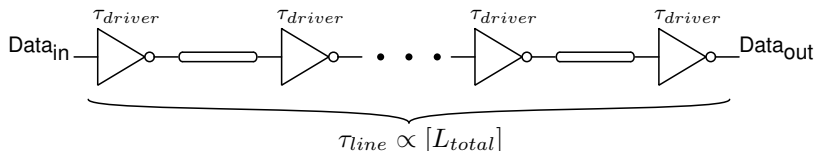


- Long interconnects, in particular, are a cause of concern.¹

¹K. C. Saraswat et al. JSSC 1982

Introduction cont'd... I

- Repeater insertion is commonly used for reducing the delay of long interconnects.¹

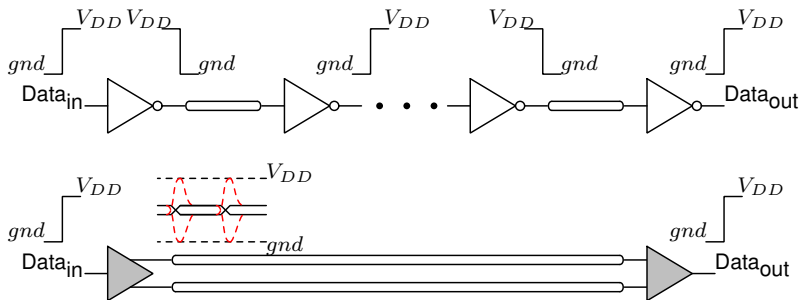


- This however
 - Comes with high power consumption
 - Increases routing complexity
 - Makes interconnect design a difficult optimization problem, involving iterative design.

¹H. B. Bakoglu et al. ISSCC 1984

Introduction cont'd... II

- These limitations of repeater inserted links have stimulated research on repeaterless low swing interconnects.¹
- Use of low swing leaves headroom available for selectively boosting high frequency components, i.e. for constructing equalizers.



¹A. Maheshwari et al. Proc. IEEE Computer Society Workshop on VLSI 2001

Introduction cont'd... III

- Many circuits have been reported for performing equalization in low swing interconnects.
- For data rates up to 2 to 5 Gbps, these have been shown to achieve
 - 10-20X reduction in power consumption over repeater inserted links (typically 100-400 fJ/bit)
 - > 3X improvement in latency
 - Greatly simplified routing

for 10 mm interconnects in CMOS technologies ranging from 180 nm to 65 nm node.

Despite these advantages, repeaterless interconnects have not been adopted in mainstream digital designs. The reasons for this form the Motivation of this work, which is stated as ...

Repeaterless interconnects have some shortcomings, which are

- 1 Long interconnects have high latency and need clock retiming circuits for sampling and this had not been well explored.
- 2 These designs use mixed signal circuits which have not been sufficiently explored for their testability, a necessary feature in most digital designs.
- 3 Unlike digital gates, these circuits are not straightforward to design. Since digital technology changes nodes quickly, any solution will be attractive only if it can keep pace with the digital processes.

These shortcomings of reported repeaterless links form the Problem statement of this work. This is stated as ...

Problem statement

We aim to design a high speed and low power repeaterless interconnect link that can be offered as a viable replacement to a repeater inserted link while meeting the following requirements

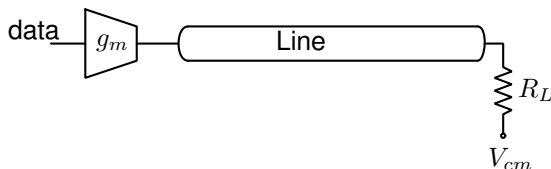
- Contain few & simple analog components to allow easy scaling
- Include synchronization so as to be an end-to-end digital link
- Feature testability with high fault coverage
- Have simple design optimization methods for quick design

- Low swing interconnect realization
 - Equalization at the Transmitter
 - Equalization at the Receiver
- Clock retiming
- Testability
- Settling time of Clock retiming circuits

- **Low swing interconnect realization**
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Low swing realization

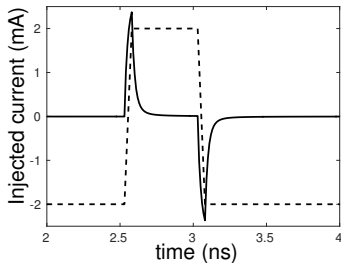
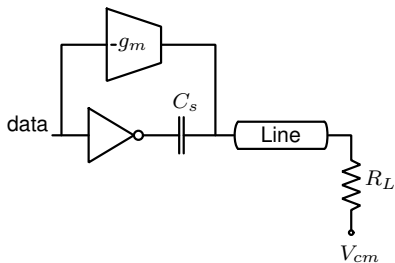
- Low swing interconnect can be implemented by using a transconductance driver with a low impedance termination at the receiver.



- Low impedance termination enhances the bandwidth of interconnect, but at the price of static power consumption.
- It is better to use a large impedance ($\sim 10\text{ k}\Omega$), and rely on equalization for bandwidth enhancement. This keeps static power low ($< 10\mu\text{W}$).

Prior art: Capacitive Equalization

- One of the ways of performing transmitter side equalization is using capacitive boosting¹.

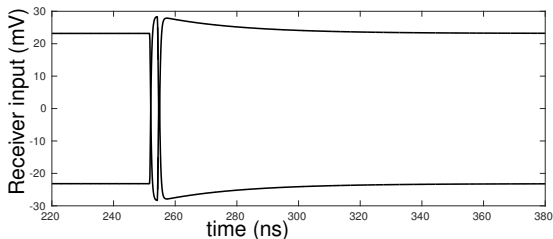
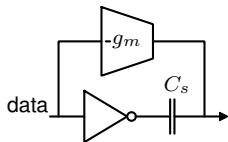


- A series capacitor injects a large current only on data transitions and hence boosts the high frequency content of the input.

¹E. Mensink et al., IEEE JSSC Feb. 2010

Time taken to reach steady state

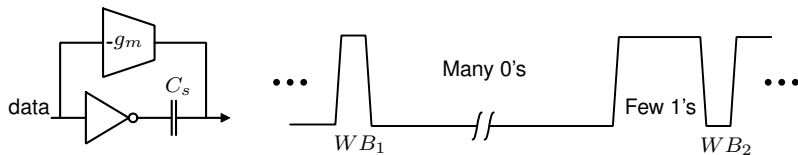
- If C_s is more than optimum, there will be overshoots
- If C_s is less than optimum, the bandwidth enhancement is poor.
- When the capacitor value is not optimum, the time taken to reach steady-state is large (can be >50 bits)



- Conventional methods of testing with PRBS sequences can be misleading, unless very long PRBS generator is used.

Prior art: Worst case sequence based design

- Dave M.¹ has reported a design optimization method for quickly designing this type of transmitters, using the interconnect response to worst case sequences.

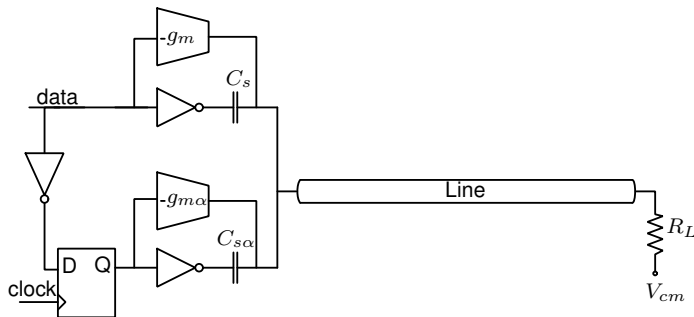


- When C_s is more than optimum, WB_2 will have least noise margin.
- When C_s is less than optimum, WB_1 will have least noise margin.
- The series capacitor value has reciprocal effects on WB_1 and WB_2 . This makes finding the optimum value very easy.

¹Dave M., PhD Thesis, IIT Bombay 2012

Capacitively coupled equalizer of 2nd order

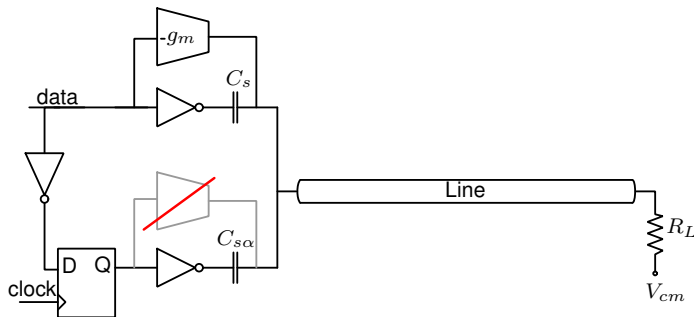
- We extended the capacitive boosting based equalizer, and constructed a second order equalizer



- A worst case sequence based design method for this second order equalizer was then formulated for finding the optimum values of g_m , C_s and $C_{s\alpha}$

Capacitively coupled equalizer of 2nd order

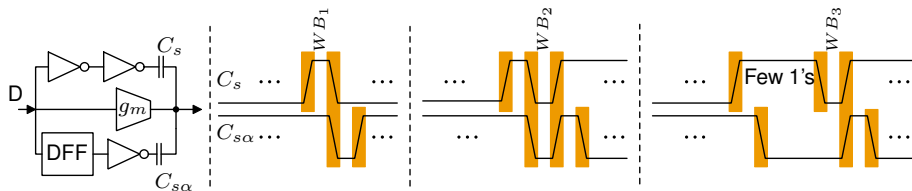
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Worst case sequences for 2nd order equalizer

- We identify three worst case sequences that pronounce the effect of each of these parameters at the receiver input.
- The first and third worst case sequences, in which the bit in question is WB_1 and WB_3 respectively, depend on $C_s - C_{s\alpha}$.



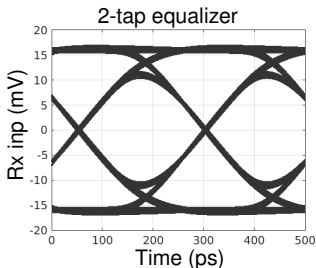
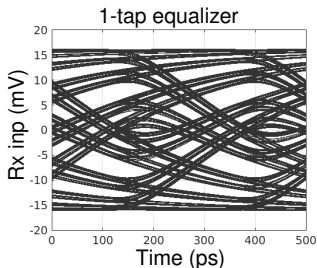
- WB_2 is driven by both C_s and $C_{s\alpha}$ simultaneously. However, it starts off from a value depending on C_s .
- The response to this bit is $\propto \frac{C_s + C_{s\alpha}}{C_s} = 1 + \frac{C_{s\alpha}}{C_s}$

Design of the transmitter

- The circuit parameters are tuned to give equal response to all three worst case sequences, which is the optimum design.
 - This results in an accurate solution, as it uses transient simulations with accurate interconnect model
 - The optimization is quick as it uses data signals with low activity factor.

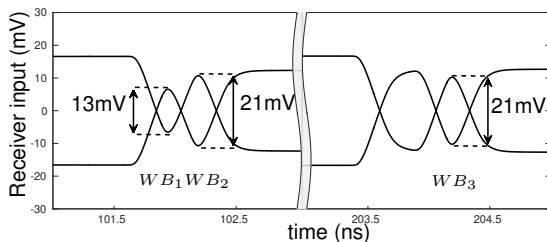
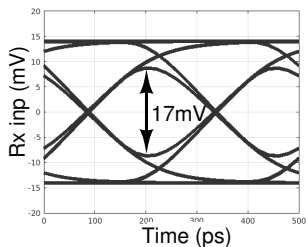
Design of the transmitter

- The circuit parameters are tuned to give equal response to all three worst case sequences, which is the optimum design.
 - This results in an accurate solution, as it uses transient simulations with accurate interconnect model
 - The optimization is quick as it uses data signals with low activity factor.
- After the design is complete, a test with random bit sequence confirms that a data signal with a good eye is received at the receiver's input.



Drawbacks of testing only with random sequence

- Testing only with random sequences can mislead to a suboptimal design, unless a very long PRBS generator is used.
- For example, here is a comparison of the response of a suboptimal design to 7-bit PRBS and worst case sequences.



Comparison with other works

We designed, laid out and performed simulation of extracted netlists of a link using this transmitter in UMC 90 nm CMOS process. Following is a comparison of the simulated results with other reported works.

Source	JSSC'08 ¹	JSSC'10 ²	JSSC'10 ³	This Work ⁴ (Simulated)
Technology	180 nm	90 nm	90 nm	90 nm
Length (mm)	10	10	10	10
Data Rate (Gbps)	1	2	4	4
Energy/bit (fJ/bit)	1000	280	371	313
Density (Gbps/ μm)	1.67	1.16	2	2
FoM = ($\frac{\text{Gbps}/\mu\text{m}}{\text{pJ/bit}}$)	1.67	4.15	5.39	6.39

¹R. Ho et al., IEEE JSSC Jan. 2008

²E. Mensink et al., IEEE JSSC Feb. 2010

³B. Kim et al., IEEE JSSC Jun. 2010

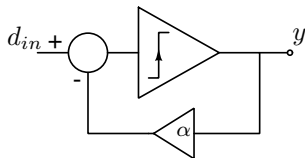
⁴Presented at VLSI Design Conference 2013

- **Low swing interconnect realization**
 - ✓ Equalization at the Transmitter
 - **Equalization at the Receiver**
- Clock retiming
- Testability
- Settling time of Clock retiming circuits

Decision feedback equalization

- Equalization at the receiver is power efficient as the equalizer circuit does not have to drive the long interconnect.
- Decision Feedback Equalization (DFE) performs *non-linear* equalization by subtracting the previous received bit's memory from the next cycle's input.

$$y[n] = \begin{cases} -1, & \text{if } d_{in}(nT) - \alpha y[n-1] < 0 \\ +1, & \text{otherwise.} \end{cases}$$



- This erases the previous bit's interference, if the decision is correct.
- The loop delay limits the maximum frequency of operation of this filter.

Decision feedback equalization cont'd...

- In practice, offset compensation has to be done for the comparator.
- The net feedback signal can then be written as

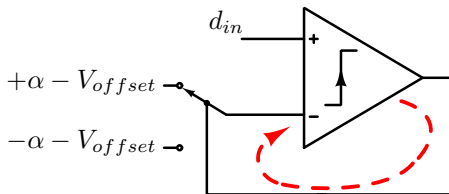
$$-\alpha y[n-1] - V_{offset} = \begin{cases} -\alpha - V_{offset}, & \text{if } y[n-1] = +1 \\ +\alpha - V_{offset}, & \text{if } y[n-1] = -1. \end{cases}$$

Decision feedback equalization cont'd...

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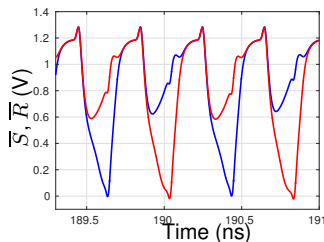
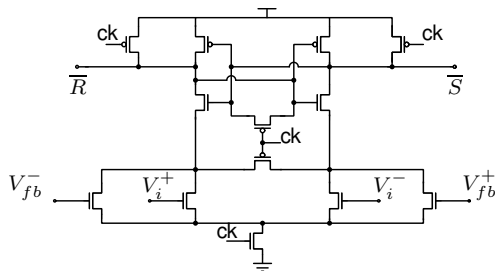
$$-\alpha y[n-1] - V_{offset} = \begin{cases} -\alpha - V_{offset}, & \text{if } y[n-1] = +1 \\ +\alpha - V_{offset}, & \text{if } y[n-1] = -1. \end{cases}$$

- We can implement this selection using switched capacitor circuits. Thus, no separate offset correction will be needed.



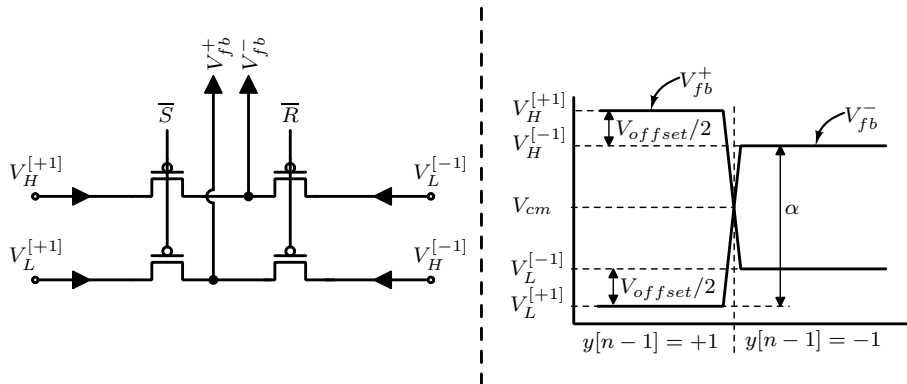
Sense amplifier comparator

- The sense amplifier comparator consists of a dynamic first stage followed by an SR slave latch.



- In order to reduce the loop latency, the feedback path is closed after the first stage, by using a switched capacitor network that is driven using the signals $\overline{S}$ and $\overline{R}$.

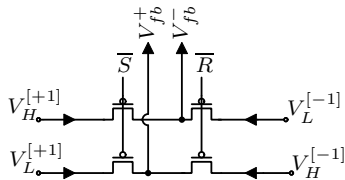
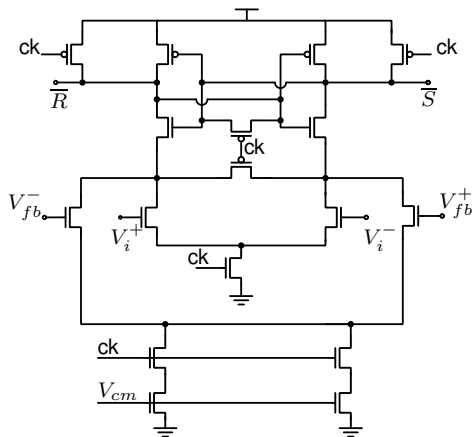
Low latency DFE circuit: Implementation



- Low V_t transistors are used for reliable and fast switching.
- This is equivalent to having a low swing dynamic latch for the decision feedback.

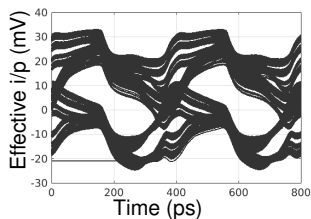
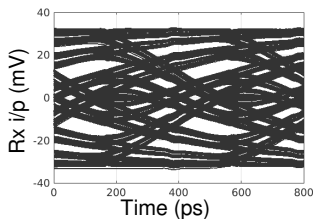
DFE implementation cont'd...

- Since $\bar{S}$ and $\bar{R}$ have large common mode swings, the feedback inputs are modified to have a high CMRR by biasing this input with a tail current source.



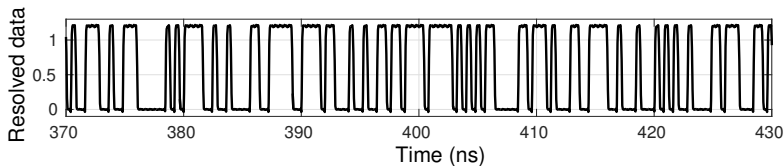
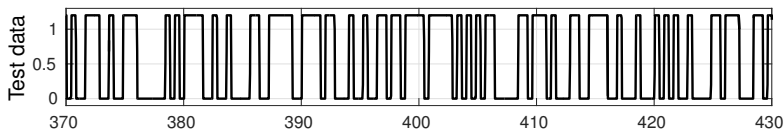
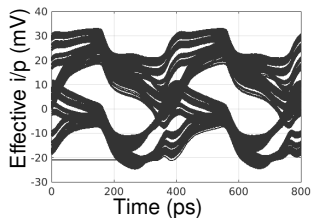
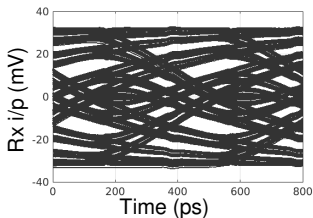
Effective input eye diagram

- The circuit was designed and tested in UMC 130 nm CMOS technology.
- A first order capacitively coupled transmitter is used with a 10 mm interconnect such that it results in a closed eye at the receiver input.
- The effective eye diagram at the input of the receiver is numerically evaluated and shown below.



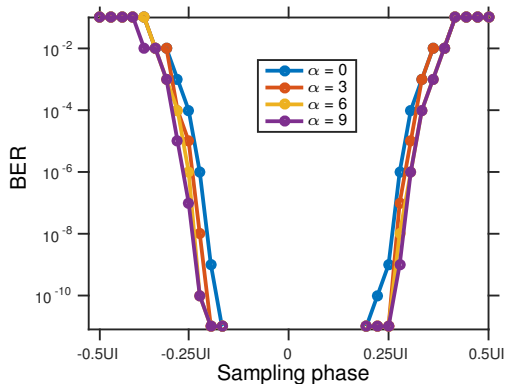
These results were obtained from post-layout simulations.

Effective input eye diagram cont'd...



Measured bath tub curves of DFE circuit

- The fabricated chip was tested at a frequency of 1 GHz¹.
- After finding and setting the offset, the DFE feedback factor was increased and the bath tub plots obtained for a few weights.²



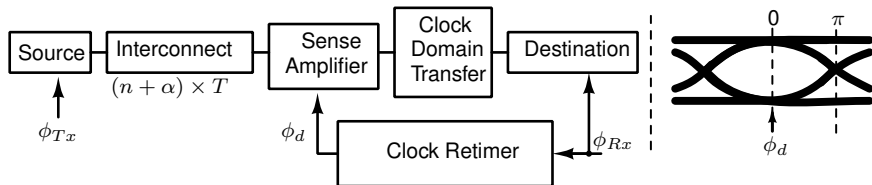
¹I thank IIT Madras for providing access to test equipment.

²This work is under review in Elsevier Microelectronics Journal

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 - ✓ Equalization at the Receiver
- **Clock retiming**
- Testability
- Settling time of Clock retiming circuits

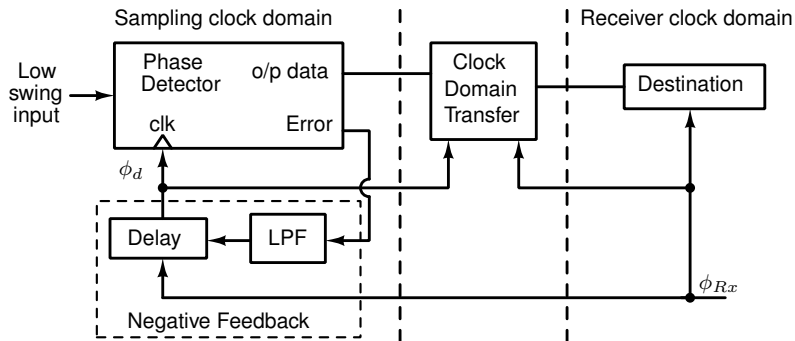
Low swing interconnect receiver

- Long interconnects have high and unknown latency.
- Due to ISI and jitter, the optimum sampling window is small.
- Clocked comparators are used for sampling because of their high speed and low power consumption.
- After the receiver samples the data and converts to CMOS levels, the data must then be handed over to receiver's clock domain.



Clock recovery circuit

- An on chip interconnect receiver has a clock running at the correct frequency available, but the correct sampling phase needs to be recovered.
- This is achieved using a phase detector to sense the phase difference between the data and the clock and a delay circuit in a negative feedback loop.



Generating delayed clock

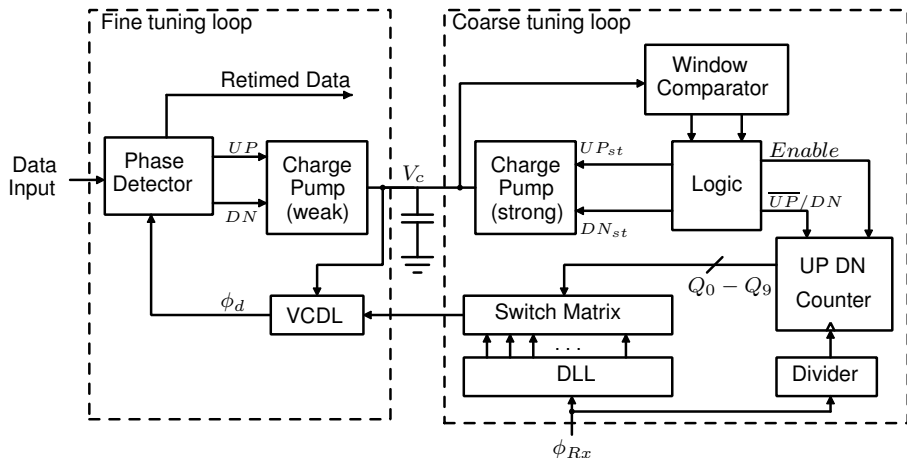
There are many ways of generating delayed clock for the clock retiming circuit.

- 1 Use a DLL to generate multiple phases and choose the appropriate one.
 - This results in phase quantization error.
- 2 Use voltage controlled delay lines.
 - Needs several stages to achieve required delay range while maintaining linearity.
- 3 Use phase interpolators.
 - Needs sinusoidal clocks of at least 4 phases which entails substantial analog circuitry.

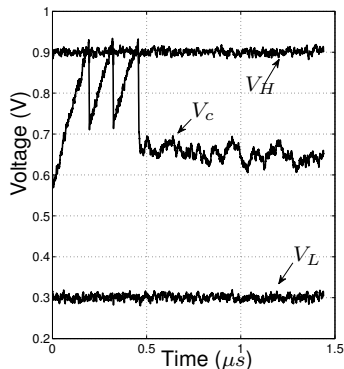
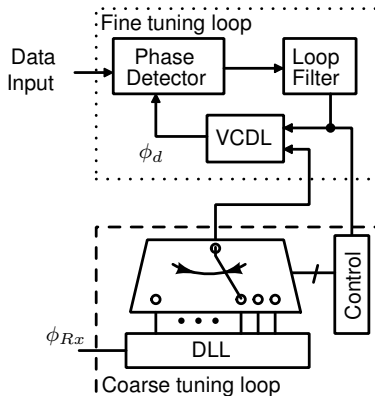
We have designed a circuit that uses a combination of 1 and 2, i.e. pick one of the phases of a DLL for a coarse correction and use a VCDL with a small range only for residual fine correction.

Coarse + fine clock synchronizer

- We use a combination of both, i.e. chose the DLL phase closest to the center of data eye and fine tune with a VCDL for accuracy.

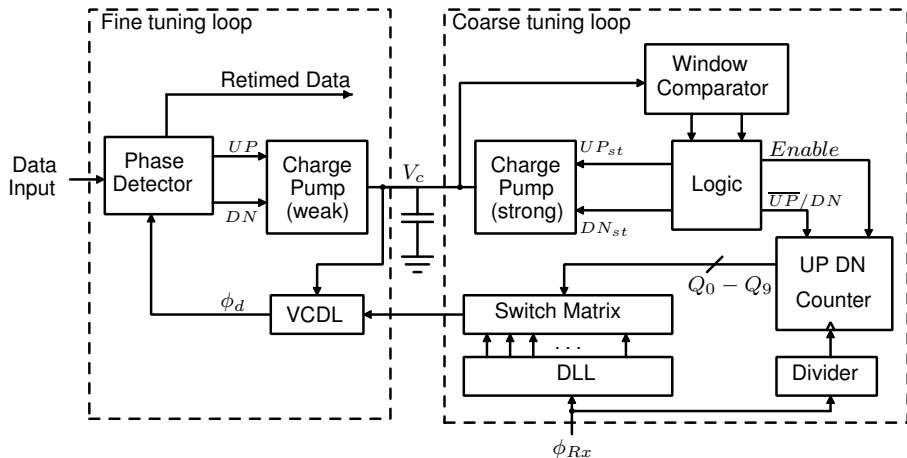


Operation of the coarse + fine clock synchronizer

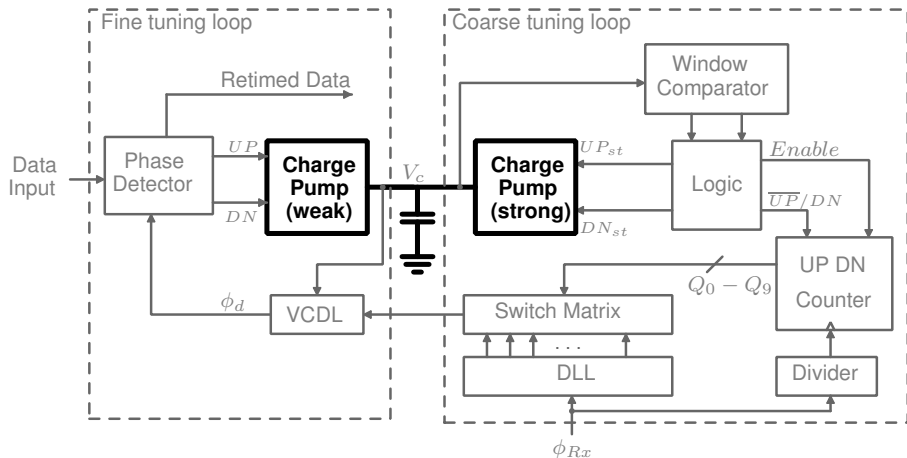


- The required VCDL range is small (≥ 1 DLL phase steps).
- The coarse tuning word can be stored and recalled for fast lock.
- Allows easy clock domain transfer (coming up ...).

Implementation: Charge Pump

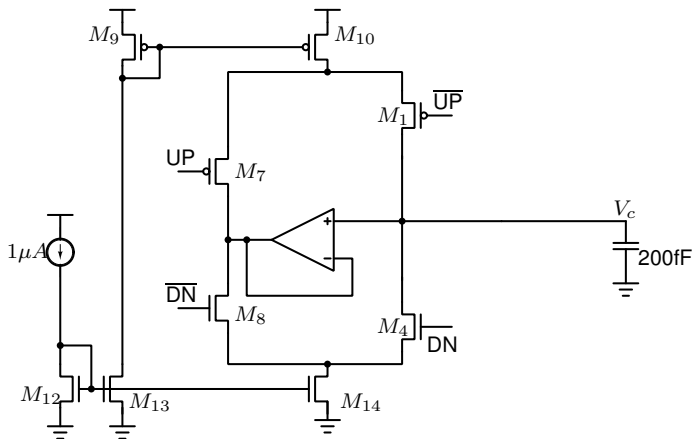


Implementation: Charge Pump



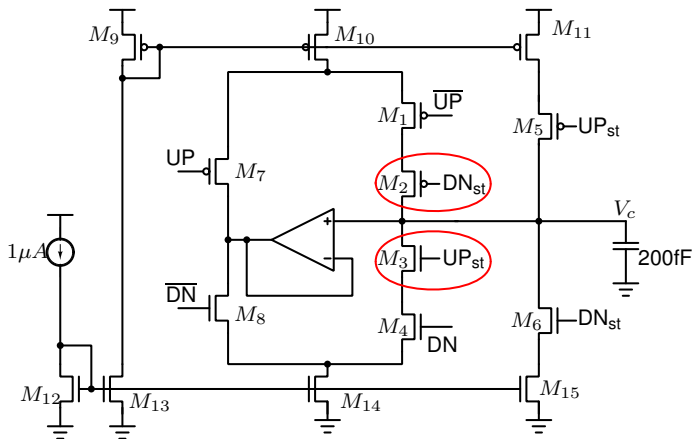
Charge pump circuit

- The loop filter capacitor is driven by two charge pumps, the weak charge pump and the strong charge pump.
- At any time only one of these charge pumps is active.



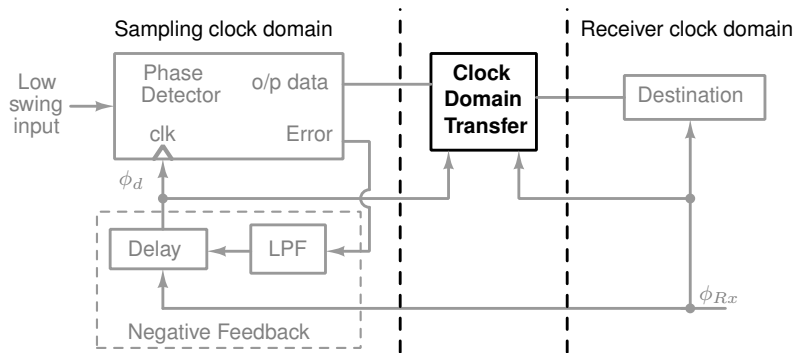
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Clock domain transfer

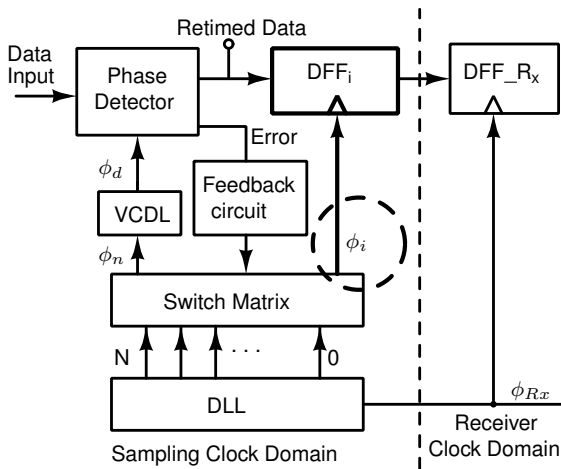
- The recovered sampling clock can occur too close to the receiver's clock phase.
- The transfer of data from the sampling clock to receiver's clock has to be done safely.



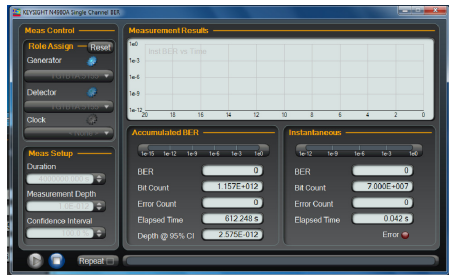
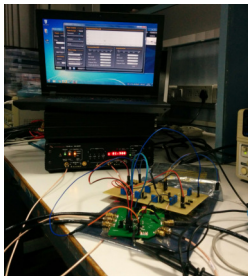
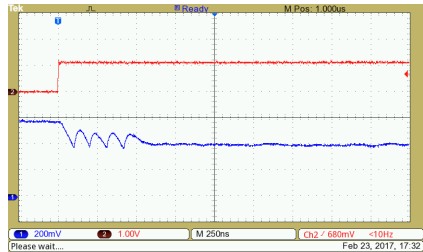
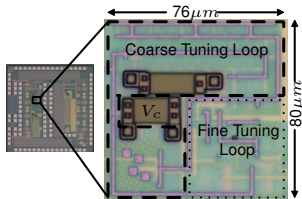
- In DLL based synchronizers, an intermediate phase can be used for performing safe clock domain transfer.

Domain crossing using intermediate DLL phase

- Once locked, the coarse tuning word indicates the correct clock phase within 2 phase steps.
- The receiver clock phase is known (the first phase of the DLL).



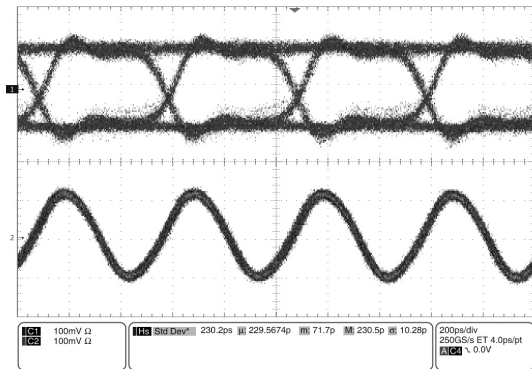
Clock retiming circuit fabricated and tested¹



¹ I thank IIT Madras for providing access to test equipment.

Retiming circuit measurements

- The chip was tested at 1.9 GHz and its BER $< 10^{-11}$.
- The rms jitter in the recovered clock is 1.6 ps.
- The retiming circuit is robust to correlated and uncorrelated jitter between the data and the clock.



Comparison with other reported works

	JSSC'14 ¹	JSSC'05 ²	JSSC'03 ³	This work ⁴	
Process	65 nm	110 nm	110 nm	130 nm (Measured)	65 nm (Simulated)
Interconnect type	On-chip	Off-chip	Off-chip	On-Chip	
Synchronizer type	Mesochronous	Plesiochronous	Plesiochronous	Mesochronous	
Control	Digital	Analog	Digital	Coarse digital + fine analog	
Delay type	Digital delay line	Phase interpolator	Phase interpolator	DLL + VCDL	
Adaptive control	X	✓	✓	✓	
Clock domain transfer	X	X	X	✓	
Data rate	3 Gb/s	10 Gb/s	10 Gb/s	1.9 Gb/s	4 Gb/s
Power consumption*	0.4 mW	220 mW	129 mW	2.9 mW	1.5 mW
Supply Voltage	0.9 V	1.5 V	1.2 V	1.2 V	1 V
Area ($\mu m \times \mu m$)	Not reported	250 x 1400	1600 x 2600	76 x 80	48 x 50

¹S.H. Lee et al., IEEE JSSC Sept. 2014

²R. Kreienkamp et al., IEEE JSSC March 2005

³H. Takauchi et al., IEEE JSSC Dec. 2003

⁴Presented at VLSI Design Conference 2017

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 - Settling time of Clock retiming circuits

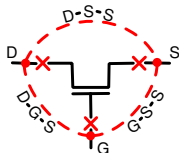
Design for Testability

- A digital circuit which is not testable is not useful for production.
- Low swing repeaterless interconnect systems that are being proposed as an alternative to repeater inserted links use mixed signal circuits.
- Unless these mixed signal circuits are testable, they cannot be used in high volume designs.
- A new design of a low swing interconnect circuit, appropriately amended to make it testable in standard digital test framework, is thus targeted.
- Digital circuits are tested using “scan tests”.

Testable low swing interconnects

- For the digital sections of the circuit, stuck-at fault model is used.
- For the analog sections, the structural fault model is used.¹ This includes the following faults

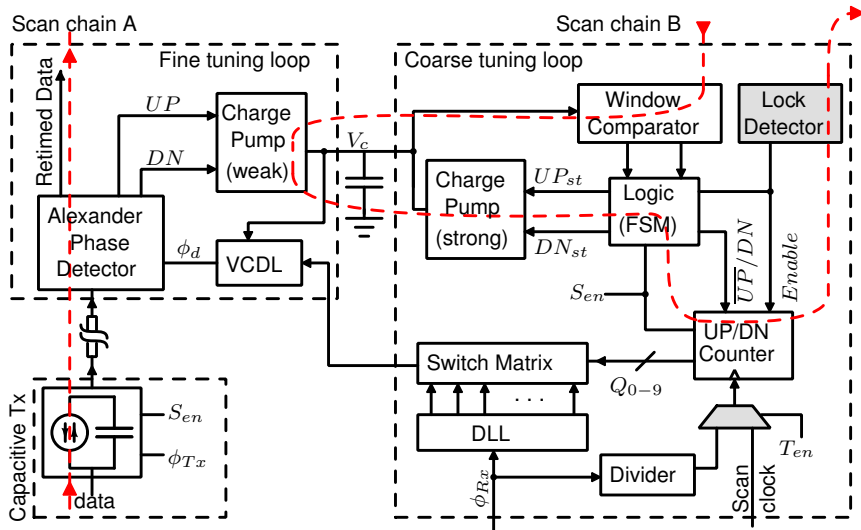
- Capacitor short
- Gate/Drain/Source : open
- Gate-Source/Gate-Drain/Drain-Source : short.



- Since the interconnect is repeaterless and has multi-cycle latency, the transmitter and receiver operate in different clock domains.
- The test of the system is divided into two sections, which are
 - The data path scan chain
 - The clock control path scan chain.

¹S. Kim et al. IEEE TCAS II, Feb. 2001

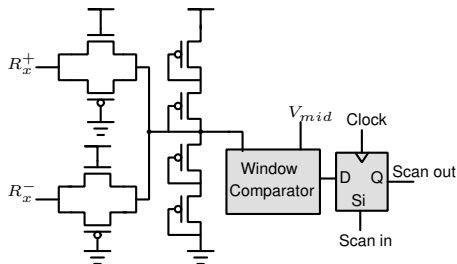
The two scan chains



Receiver termination

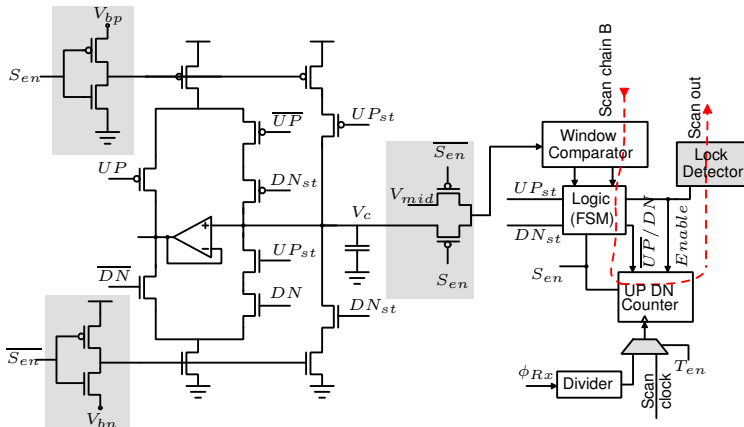
- A fault free circuit is balanced and the common mode at the receiver is static.
- Any fault in the transmitter or the receiver results in a dynamic mismatch that results in the receiver common mode to swing with the data.
- A window comparator sensing this node can detect most faults in the transmitter at scan frequencies.

- The window comparator is pre-programmed for a window of 30 mV.



Clock control: Testing the charge pump

- During scan test the charge pump's current sources are converted to on switches, making it a combinational circuit.
- The output of this circuit depends on the value pre-loaded into scan chain A and affects the output of scan chain B.



Fault coverage

- Using scan test, by converting charge pump to combinational circuit, 75% of the faults are detected.
- A simple BIST increases the fault coverage to 95% of the faults.

Defect	Coverage
Gate open	87.8%
Drain open	93.9%
Source open	93.9%
Gate drain short	93.9%
Gate source short	100%
Drain source short	100%
Capacitor short	100%
Total	94.8%

Entity	Number
Flip-flop	7
Comparators (DC)	4
Comparators (100 MHz)	2
D-Latch	1
2×1 Multiplexer	2
3 bit saturating UP counter	1
Control signals	2
Logic gates	6
Area	29%

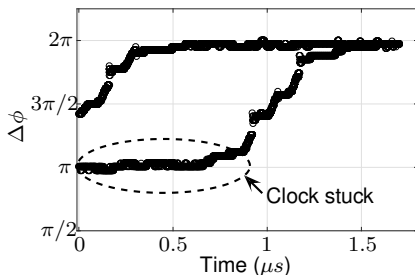
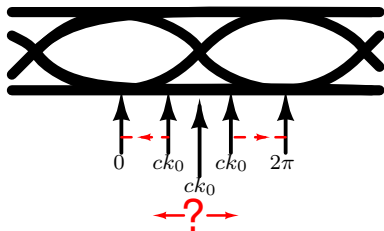
- These tests can be performed using existing test infrastructure meant for testing digital circuits.¹

¹Presented at DATE Conference 2016

- ✓ Low swing interconnect realization
 - ✓ Equalization at the Transmitter
 - ✓ Equalization at the Receiver
- ✓ Clock retiming
- ✓ Testability
- **Settling time of Clock retiming circuits**

Mesochronous Systems- Settling Time

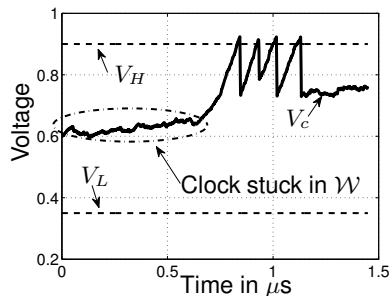
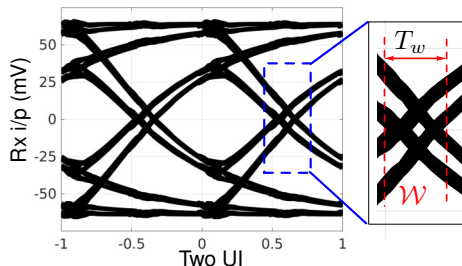
- The time to acquire lock i.e. Settling time - depends on initial phase error $\Delta\phi$
- If initial phase error $< \pi$: $\Delta\phi$ is reduced to 0
- If initial phase error $> \pi$: $\Delta\phi$ is increased to 2π ($= 0$ by wrapping)



- But, if initial phase error $\sim \pi$, system may remain stuck for extended period of time!

Window of Susceptibility $\mathcal{W}$

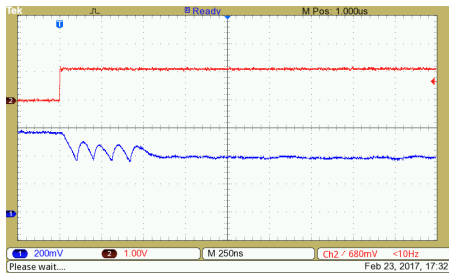
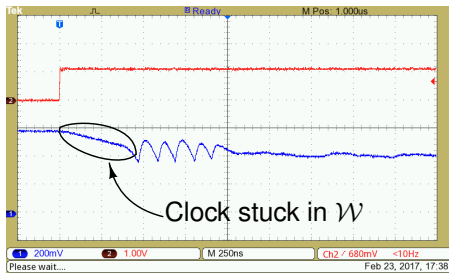
- This extension of settling time is due to jitter in the data.
- We define a Window of susceptibility $\mathcal{W}$ whose width T_w = width of horizontally closed region of data eye



- If initial clock is in $\mathcal{W}$, settling time is non-deterministic

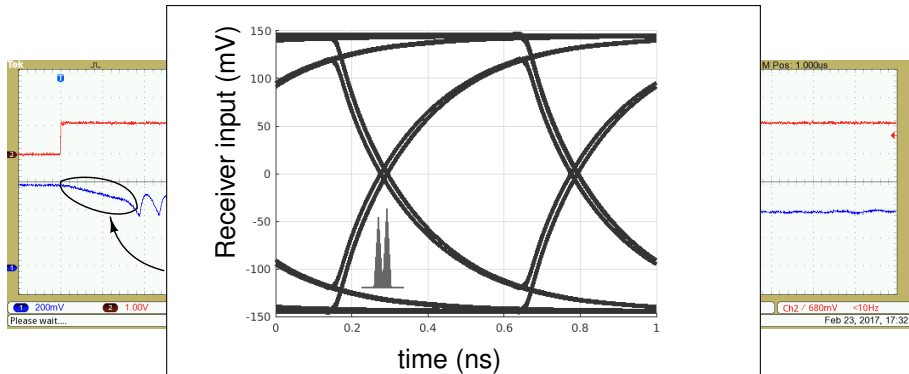
Measured extension in settling time

- The dependence of settling time on the jitter in the data was demonstrated using the fabricated circuit of the retiming circuit.
- Measurement done at 2 GHz. Clock stuck for ~ 725 cycles when initialized in the center of the window.



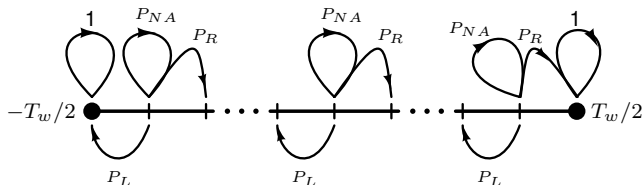
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Markov model of the synchronizer

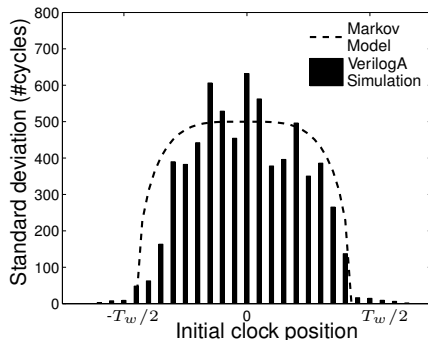
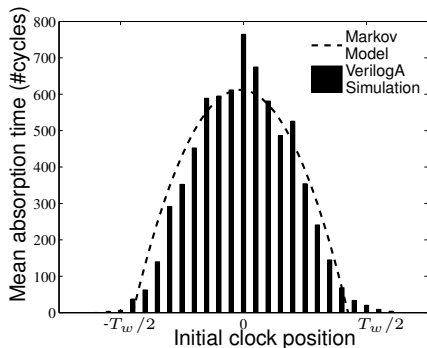
- To analyze the settling time we have constructed a Markov chain model of the system.
- The clock position is quantized to the step size of the fine control loop, and each position is modeled as a state of the Markov chain.



- Once the clock reaches the edge of the window, the circuit locks to center of eye monotonically.
- Knowing the characteristics of the jitter, the jump probabilities can be calculated.

Analysis example : Jitter due to 1 bit ISI

- The Markov chain was constructed to model the scenario when the jitter is due to ISI limited to 1 previous bit.
- The mean time to absorption, which is representative to the settling time is evaluated using the model and also using behavioural simulations.

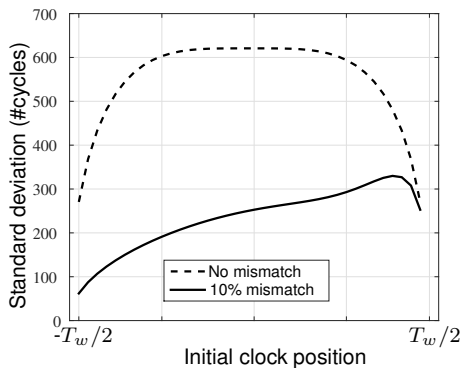
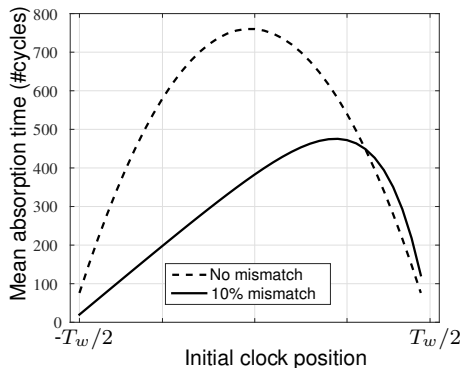


Further applications of the Markov model

- The Markov chain model can be used to analyze different types of jitter. We have used it to analyze
 - data dependent (ISI induced) jitter of only previous bit,
 - data dependent (ISI induced) jitter of previous 2 bits,
 - random clock jitter,
 - simultaneous data dependent and random jitter.
- Since the settling time is probabilistic, it can theoretically extend to infinity.
- However, for a given confidence level we can predict the worst case settling time by analyzing the Markov model.
- We have also used the model to suggest designs that have reduced settling time, like
 - introducing mismatch in left and right steps,
 - reversing the order of coarse and fine correction,which were then verified with circuit simulations.

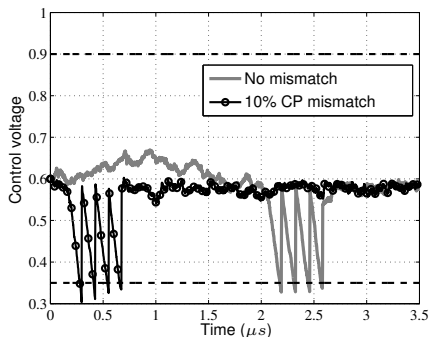
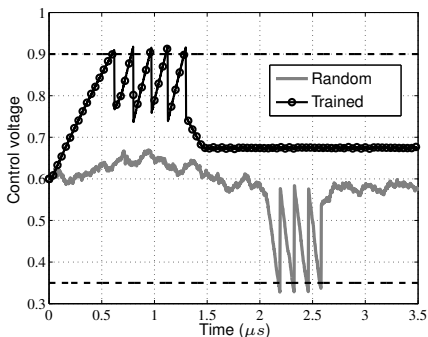
Effect of UP and DN mismatch

- A 10% mismatch in the UP/DN weights reduces the mean absorption time by almost 40%.
- This essentially biases the system towards one of the edges.
- Makes settling time more predictable as well.



Reduction of settling time with UP/DN pump mismatch

- This can be implemented by introducing a mismatch in the relative strengths of UP and DN currents in the charge pump.
- A training sequence, that is designed to have asymmetrical ISI can also produce such mismatch.



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Conclusions

- In scaled technologies, repeater inserted long interconnects limit the performance of digital chips.
- Repeaterless interconnects have been proposed in the literature as an alternative approach to repeater insertion and have been in discussion for almost 15 years.
- These repeaterless schemes have however had some limitations, like
 - Clock retiming for sampling at optimum instant had not received enough attention.
 - Testability of these systems, which use mixed signal circuits, was not investigated sufficiently.
 - Fast design techniques for quickly designing these schemes for different interconnect lengths and in different technologies were not sufficiently explored.

Conclusions : Thesis contributions

- ✓ Analyzed transmitter side equalization using capacitive boosting.
 - Designed a second order capacitive equalizer with a weak driver.
 - Developed a design method for quickly designing this transmitter using worst case sequences.
- ✓ Designed a decision feedback equalizer, using a sense amplifier comparator $\Rightarrow$ has low latency and includes offset compensation.
- ✓ Proposed a clock retiming circuit for sampling low swing data.
 - The circuit uses a coarse + fine tuning with background tracking.
 - Clock domain transfer is included in the design.
- ✓ Demonstrated how repeaterless interconnects can be tested within the existing test infrastructure.
- ✓ Analyzed the influence of timing jitter on the settling time of mesochronous clock retiming circuits.
 - Developed a Markov model for analyzing different types of jitter.
 - Designed fast settling synchronizers and verified them.

Future directions of research in this area

- Design of low swing interconnect bus
 - Worst case sequence based design for crosstalk correction?
 - Clock retiming circuits $\Rightarrow$ common coarse control with individual fine control?
- DVFS of low swing interconnects.
- Automatic test pattern generation for mixed signal circuits.
 - Automatic evaluation of fault coverage of structural faults.

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- DVFS of low swing interconnects.
- Automatic test pattern generation for mixed signal circuits.
 - Automatic evaluation of fault coverage of structural faults.
- **Automatic synthesis of digital chips with low swing interconnects.**
 - **Initially use a library of transceivers for different lengths?**
 - **Design of algorithms for automatic synthesis?**

Thank You.